

Def. A function is continuous at a if:

- $\lim_{x \rightarrow a} f(x) = f(a)$
- $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

whenever $0 < |x - a| < \delta$,

then $|f(x) - f(a)| < \varepsilon$.

Ex. If f is continuous at a , then show that $\forall \varepsilon > 0, \exists \delta > 0$, s.t.

$$|x - a| < \delta,$$

$$|y - a| < \delta$$

then $|f(x) - f(y)| < \varepsilon$.

Rough Work

$$\begin{aligned} & |f(x) - f(a) + f(a) - f(y)| \\ & \leq \underbrace{|f(x) - f(a)|}_{\frac{\varepsilon}{2}} + \underbrace{|f(a) - f(y)|}_{\frac{\varepsilon}{2}} < \varepsilon \end{aligned}$$

Pf. let $\varepsilon > 0$

Since f is cts. at a , $\exists \delta > 0$, s.t.

$|z - a| < \delta$, we have

$$|f(z) - f(a)| < \frac{\varepsilon}{2}$$

$\Rightarrow$ For $z = x, y$, if $|x - a| < \delta$
& $|y - a| < \delta$

$$|f(x) - f(a)| < \frac{\varepsilon}{2}$$

$$|f(y) - f(a)| < \frac{\varepsilon}{2}$$

By $\triangle$ ineq,

$$|f(x) + f(y)| \leq |f(x) - f(a)| + |f(a) - f(y)|$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon.$$



Def. A property P on a function f is an open property if P holds at a , then there exists an open interval containing a on which P is true everywhere.

Thm. Being positive at point, a is an open property for continuous functions.
(i.e. If f continuous and $f(a) > 0$, then $\exists$ some open interval I containing a s.t. $\forall x \in I, f(x) > 0$).

PF Take $\varepsilon = \frac{f(a)}{2}$.

Since $f(a) > 0, \varepsilon > 0$

Then for this $\varepsilon, \exists \delta > 0$, s.t.

whenever $0 < |x - a| < \delta$,

we have $|f(x) - f(a)| < \varepsilon = \frac{f(a)}{2}$.

We claim that we've found the interval I .

Let $I = (a - \delta, a + \delta)$.

Then $a \in I$.

For any $x \in I$, we also have

$$f(a) - \varepsilon < f(x) < f(a) + \varepsilon$$

In particular,

$$\begin{aligned} f(x) &> f(a) - \varepsilon \\ &> f(a) - \frac{f(a)}{2} \\ &= \frac{f(a)}{2} > 0 \end{aligned}$$

So for $\forall x \in I$, $f(x) > 0$.

Then, positivity is an open interval.



Ex. Continuity at a point a is not an open property

Counterexample

The function $f(x) = \begin{cases} 1/q & \text{if } x = p/q \text{ in lowest terms} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$

is continuous only at irrational x .

Since $\mathbb{Q}$ is dense in $\mathbb{R}$ in any open I (containing $\sqrt{2}$)

We can find $p/q \in I$.

$f(x)$ is not cts at p/q

$\Rightarrow$ Continuous is not an open property.



Def. A property P on a function f at a point a is a local property if for any open interval I with $a \in I$ and any function g with $f = g$ on I , then P also holds for g at a .

i.e. The property depends only on a small neighborhood around a , so the rest doesn't matter.

Ex. ① The property of the point a being a local maximum for a function f is a local property.

② The property of being the global maximum is not a local property.

③ $\lim_{x \rightarrow a} f(x) = l$

By def, $\forall \varepsilon > 0, \exists \delta > 0$ s.t.
 $0 < |x - a| < \delta$

$$|f(x) - l| < \varepsilon$$

$$\Rightarrow \lim_{x \rightarrow a} f(x) = l$$

is a local property.

 this is the I you want to consider, whatever f does outside of I does not influence the limit.

④ f being cts at a is a local property.

⑤ Positivity is a local property

Exercise. Show that if f is cts, then $|f|$ is also cts.

PF. Since $|f| = | \cdot | \circ f$ and f cts, just need to show $| \cdot |$ is cts.

Since continuity is a local property

$$\text{If } I \neq 0, \text{ then } | \cdot | = \begin{cases} x \\ -x \end{cases}$$

So $| \cdot |$ cts. at $\forall \mathbb{R} \setminus \{0\}$.

Exercise Use ϵ - δ to show continuity at 0.